

PREDICTIVE ANALYTICS FOR HEALTHCARE: UTILIZING REGRESSION MODELS TO FORECAST PATIENT LENGTH OF STAY IN URBAN HOSPITALS – YOUR TITLE

**Project Report Submitted in fulfillment of the
Requirements for the Award of the Internship of
Summer Training Program by IDRA**

Data Science and AI

By:

Name: [Student Name]

Institute: [Institute Name]

Institute Roll No.: [Roll Number]

Enrollment No.: IDRA-[Enrollment Number]

Under the Supervision of

Dr. Shaheena Salam

Through the platform:



India Data Research Academy

DECLARATION

Replace this placeholder with a brief declaration stating that the work presented in the report is your own, that the analysis and code were completed by you, and that all external sources have been properly acknowledged.

ACKNOWLEDGEMENT

Write a short acknowledgement thanking the IDRA training program, supervisor, institute, dataset/source providers, and anyone who directly supported the project.

ABSTRACT

Write a concise 150-250 word summary covering the problem, dataset, methodology, major analytical findings, model used, model performance, and the main conclusion. Do not include citations, code, or detailed tables here.

Keywords: [Keyword 1]; [Keyword 2]; [Keyword 3]; [Keyword 4]; [Keyword 5]

TABLE OF CONTENTS

1. Introduction
 2. Problem Statement and Motivation
 3. Objectives
 4. Research Questions
 5. Dataset Description and Data Understanding
 6. Data Cleaning
 7. Data Preprocessing
 8. Exploratory Data Analysis
 9. Statistical Analysis
 10. Feature Engineering
 11. Model Development
 12. Model Evaluation
 13. Results and Findings
 14. Discussion
 15. Limitations
 16. Conclusion
 17. Recommendations and Future Work
 18. References
- Appendix A. Code
- Appendix B. Additional Figures and Tables

After completing the report in Microsoft Word, update the Table of Contents so that the page numbers correspond to the final document. Keep the hierarchy of numbered headings consistent throughout the report.

1. INTRODUCTION

1.1 Background

Introduce the domain of your project and explain why the problem matters. Give enough context for a reader who is not familiar with the dataset or domain.

1.2 Context of the Problem

Explain the real-world context, stakeholders, and practical situation in which the analysis or prediction would be used.

1.3 Scope of the Study

Define what your project covers and, just as importantly, what it does not cover.

2. PROBLEM STATEMENT AND MOTIVATION

State the problem as a clear, specific question. Explain the motivation for solving it and what value a data-driven solution could provide.

3. OBJECTIVES

List 3-6 measurable objectives. Each objective should describe an outcome of the project, not merely a software action.

- To understand and prepare the selected dataset for analysis.
- To identify important patterns, relationships, and factors within the data.
- To build a machine learning model to predict [target variable].
- To evaluate the model using appropriate performance metrics.
- To translate the results into meaningful conclusions and recommendations.

Replace or edit these sample objectives so they match your own project.

4. RESEARCH QUESTIONS

Write 3-6 questions that your EDA and modelling work will answer. Your questions should guide the analysis rather than simply describe what code you will run.

1. What are the major characteristics and quality issues in the dataset?
2. Which variables are most strongly associated with the target outcome?
3. How does the target vary across important groups or categories?
4. How accurately can the selected model predict the target outcome?

5. DATASET DESCRIPTION AND DATA UNDERSTANDING

5.1 Data Source

State where the dataset came from, the access date if relevant, and any licensing or usage information. If the dataset was provided in the LMS, mention that clearly.

5.2 Dataset Size and Structure

Report the number of rows, columns, and the role of each important variable. Identify the target variable and explain why it is the target.

Column 1	Column 2	Column 3	Column 4
----------	----------	----------	----------

[Enter value]	[Enter value]	[Enter value]	[Enter value]
[Enter value]	[Enter value]	[Enter value]	[Enter value]
[Enter value]	[Enter value]	[Enter value]	[Enter value]
[Enter value]	[Enter value]	[Enter value]	[Enter value]
[Enter value]	[Enter value]	[Enter value]	[Enter value]

Table 1. Summary of the Dataset

Replace the sample table with your own results. Mention the table in the text before or after it.

5.3 Data Dictionary

Create a compact data dictionary. Include variable name, meaning, data type, and role (feature/target) for the main variables.

Column 1	Column 2	Column 3	Column 4
[Enter value]	[Enter value]	[Enter value]	[Enter value]
[Enter value]	[Enter value]	[Enter value]	[Enter value]
[Enter value]	[Enter value]	[Enter value]	[Enter value]
[Enter value]	[Enter value]	[Enter value]	[Enter value]
[Enter value]	[Enter value]	[Enter value]	[Enter value]

Table 2. Data Dictionary

Replace the sample table with your own results. Mention the table in the text before or after it.

6. DATA CLEANING

Document the state of the raw data before cleaning. Explain missing values, duplicates, invalid values, inconsistent categories, incorrect data types, and outliers if present. For every important cleaning decision, explain why you chose that treatment.

6.1 Missing Values

Report which columns contain missing values, how many are missing, what proportion this represents, and how you handled them.

6.2 Duplicate Records

State whether duplicates were found, how many were removed, and whether duplicate removal changed the dataset size.

6.3 Data Types and Inconsistent Values

Describe any data type corrections or inconsistent text/category values you fixed.

6.4 Outlier Handling

Explain which variables were checked, which method was used, what was detected, and what decision was made. Do not remove observations without justification.

6.5 Before vs After Cleaning

Column 1	Column 2	Column 3	Column 4
[Enter value]	[Enter value]	[Enter value]	[Enter value]
[Enter value]	[Enter value]	[Enter value]	[Enter value]

[Enter value]	[Enter value]	[Enter value]	[Enter value]
[Enter value]	[Enter value]	[Enter value]	[Enter value]

Table 3. Data Quality Before and After Cleaning

Replace the sample table with your own results. Mention the table in the text before or after it.

7. DATA PREPROCESSING

Explain how the cleaned dataset was prepared for analysis and modelling. Include encoding, scaling, transformations, feature selection, and train-test splitting where applicable. Make it clear which steps were performed before and after the train-test split.

7.1 Categorical Encoding

Identify categorical features and explain whether you used one-hot, ordinal, label encoding, or another method, and why.

7.2 Feature Scaling and Transformation

Explain which numerical features were scaled or transformed and why that was appropriate for the selected model.

7.3 Train-Test Split

State the train-test split ratio, random state if used, and explain why the split is necessary. Mention how you avoided data leakage.

7.4 Final Feature Set

List the final input features and target variable used for modelling.

8. EXPLORATORY DATA ANALYSIS

Use EDA to understand distributions, group differences, trends, and relationships. Do not add charts only because they were taught in class; each chart should answer a meaningful analytical question.

8.1 Univariate Analysis

```
# Paste your relevant code here
```



Figure 1. Distribution of [Important Variable]

Insert your chart/graph here. Keep the figure centered and readable.

Explain what the distribution shows, whether it is skewed or approximately symmetric, and whether there are unusual observations.

8.2 Categorical Analysis

```
# Paste your relevant code here
```

[INSERT FIGURE 2 HERE]

Figure 2. [Category] Compared Across [Outcome/Measure]

Insert your chart/graph here. Keep the figure centered and readable.

Explain the key difference between categories and why it matters.

8.3 Bivariate / Relationship Analysis

```
# Paste your relevant code here
```

[INSERT FIGURE 3 HERE]

Figure 3. Relationship Between [Feature] and [Target]

Insert your chart/graph here. Keep the figure centered and readable.

Explain the direction, strength, and practical meaning of the relationship. Do not claim causation from correlation alone.

8.4 Correlation Analysis

```
# Paste your relevant code here
```

[INSERT FIGURE 4 HERE]

Figure 4. Correlation Heatmap of Numerical Variables

Insert your chart/graph here. Keep the figure centered and readable.

Identify important positive or negative relationships and discuss whether they have implications for modelling.

8.5 Other Figures

```
# Paste your relevant code here
```

9. STATISTICAL ANALYSIS

Report relevant descriptive statistics and statistical observations. Depending on your project, include mean, median, mode, standard deviation, variance, IQR, group-level summaries, or other statistics covered during the training.

Column 1	Column 2	Column 3	Column 4
[Enter value]	[Enter value]	[Enter value]	[Enter value]
[Enter value]	[Enter value]	[Enter value]	[Enter value]
[Enter value]	[Enter value]	[Enter value]	[Enter value]
[Enter value]	[Enter value]	[Enter value]	[Enter value]
[Enter value]	[Enter value]	[Enter value]	[Enter value]

Table 4. Descriptive Statistics of Key Variables

Replace the sample table with your own results. Mention the table in the text before or after it.

Interpret the statistics in words. Avoid listing numbers without explaining what they indicate.

10. FEATURE ENGINEERING

Describe any new variables created from the original data. Explain the logic behind each engineered feature and how it may help answer the research question or improve prediction.

Column 1	Column 2	Column 3	Column 4
[Enter value]	[Enter value]	[Enter value]	[Enter value]
[Enter value]	[Enter value]	[Enter value]	[Enter value]
[Enter value]	[Enter value]	[Enter value]	[Enter value]
[Enter value]	[Enter value]	[Enter value]	[Enter value]

Table 5. Engineered Features

Replace the sample table with your own results. Mention the table in the text before or after it.

11. MODEL DEVELOPMENT

11.1 Problem Formulation

Clearly state whether your task is regression or classification. Define the input features and target variable. Explain why the chosen problem formulation makes sense.

11.2 Model Selection

Explain why the chosen model is appropriate. Mention any alternative model you considered, if applicable.

11.3 Training the Model

Include the relevant model-training code in the notebook/report and explain the main parameters. Keep code focused on the important steps. Full code can also be placed in Appendix A.

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)
```

The code above is only a formatting example. Replace it with the code for your own project.

12. MODEL EVALUATION

Evaluate the trained model on unseen test data. Select metrics appropriate to the problem type and explain what each metric means. Report results clearly and interpret them rather than merely printing metric values.

12.1 Regression Evaluation

If your project is regression, report the regression metrics covered in training, such as MAE, MSE, RMSE, and R-squared, where appropriate.

Column 1	Column 2	Column 3	Column 4
[Enter value]	[Enter value]	[Enter value]	[Enter value]
[Enter value]	[Enter value]	[Enter value]	[Enter value]
[Enter value]	[Enter value]	[Enter value]	[Enter value]
[Enter value]	[Enter value]	[Enter value]	[Enter value]

Table 6. Regression Model Performance

Replace the sample table with your own results. Mention the table in the text before or after it.

12.2 Classification Evaluation

If your project is classification, report the confusion matrix, accuracy, precision, recall, and F1-score as appropriate.



Figure 5. Confusion Matrix

Insert your chart/graph here. Keep the figure centered and readable.

Column 1	Column 2	Column 3	Column 4
[Enter value]	[Enter value]	[Enter value]	[Enter value]
[Enter value]	[Enter value]	[Enter value]	[Enter value]
[Enter value]	[Enter value]	[Enter value]	[Enter value]
[Enter value]	[Enter value]	[Enter value]	[Enter value]

Table 7. Classification Model Performance

Replace the sample table with your own results. Mention the table in the text before or after it.

12.3 Overfitting and Underfitting

Compare training and test performance. Explain whether the model appears well-fitted, overfitted, or underfitted, and support your conclusion with evidence.

13. RESULTS AND FINDINGS

Bring together the most important results from the EDA and model. Highlight the findings that directly answer your research questions.

1. Key Finding 1: [State the finding, cite the supporting figure/table/result, and explain its practical meaning.]
2. Key Finding 2: [State the finding, cite the supporting figure/table/result, and explain its practical meaning.]

3. Key Finding 3: [State the finding, cite the supporting figure/table/result, and explain its practical meaning.]
4. Key Finding 4: [State the finding, cite the supporting figure/table/result, and explain its practical meaning.]
5. Key Finding 5: [State the finding, cite the supporting figure/table/result, and explain its practical meaning.]

14. DISCUSSION

Interpret the overall results in context. Discuss why the findings may have occurred, how they relate to the problem, and what a practitioner or decision-maker should take away from them. Distinguish observed relationships from causal claims.

15. LIMITATIONS

Be honest about data, methodological, modelling, and scope limitations. Examples include sample bias, missing variables, limited data volume, measurement quality, assumptions of the model, or limits on generalization.

- [Limitation 1]
- [Limitation 2]
- [Limitation 3]

16. CONCLUSION

Provide a concise conclusion that returns to the original problem and objectives. Summarize what the analysis established, how useful the model was, and what the overall project demonstrates. Do not introduce new evidence here.

17. RECOMMENDATIONS AND FUTURE WORK

17.1 Recommendations

Provide 3-5 practical recommendations based directly on your findings. Each recommendation should be specific enough that a stakeholder could act on it.

- Recommendation 1: [Action + reason + evidence from your analysis]
- Recommendation 2: [Action + reason + evidence from your analysis]
- Recommendation 3: [Action + reason + evidence from your analysis]

17.2 Future Work

Explain what could be improved or extended in a future study, such as additional data, alternative models, new variables, better sampling, deployment, or more rigorous validation.

18. REFERENCES

List every external source used in the report, including dataset sources, papers, documentation, articles, and websites. Use one consistent citation style throughout the report. Do not include sources that were not actually used.

[Author/Organization]. (Year). [Title]. [Publisher/Website]. [URL if applicable].

[Second reference]

[Third reference]

APPENDIX A. CODE

Place important code needed to reproduce your analysis here. Keep the main report readable by showing only key code in the methodology/model sections; place longer code listings in the appendix. Use clear subheadings for data loading, cleaning, preprocessing, EDA, and modelling.

A.1 Data Loading and Inspection

Paste your relevant code here

A.2 Data Cleaning

Paste your relevant code here

A.3 Preprocessing

Paste your relevant code here

A.4 EDA and Visualization

Paste your relevant code here

A.5 Model Training and Evaluation

Paste your relevant code here

APPENDIX B. ADDITIONAL FIGURES AND TABLES

Use this section for additional charts, tables, model outputs, or supplementary results that are useful but not essential to the main narrative.



Figure 6. Additional Figure

Insert your chart/graph here. Keep the figure centered and readable.

Column 1	Column 2	Column 3	Column 4
[Enter value]	[Enter value]	[Enter value]	[Enter value]
[Enter value]	[Enter value]	[Enter value]	[Enter value]
[Enter value]	[Enter value]	[Enter value]	[Enter value]
[Enter value]	[Enter value]	[Enter value]	[Enter value]

Table 8. Additional Results Table

Replace the sample table with your own results. Mention the table in the text before or after it.

